FA 4: Dummy-Variable Regression Using Real-World Data

2026-09-29

Part A. Exploring and Preparing the Data

1. How many observations and variables are contained in the dataset?

The `diamonds` dataset contains 53,940 observations and 10 variables.

2. Identify the response variable and explain what it measures.

The response variable is price. It measures the price of the diamond in US dollars.

```
summary(diamonds$price)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     326     950    2401    3933    5324   18823
```

3. Identify the quantitative explanatory variable used in this assessment and describe its measurement.

The quantitative explanatory variable is carat. It measures the weight of the diamond in carats.

```
summary(diamonds$carat)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.2000  0.4000  0.7000  0.7979  1.0400  5.0100
```

4. Identify the categorical explanatory variable and list its five categories.

The categorical explanatory variable is cut. Its five categories are Fair, Good, Very Good, Premium, and Ideal.

```
levels(diamonds$cut)
```

```
## [1] "Ideal"     "Fair"      "Good"      "Premium"   "Very Good"
```

```
table(diamonds$cut)
```

```
##
##     Ideal      Fair      Good   Premium Very Good
##     21551      1610      4906     13791     12082
```

5. Select one category of cut as the reference category and explain why a reference category is necessary when using dummy variables.

The reference category is Ideal. A reference category is necessary so that the other categories can be compared with one baseline category and to avoid perfect multicollinearity when an intercept is included.

```r
levels(diamonds$cut)
```

```
## [1] "Ideal"     "Fair"      "Good"      "Premium"   "Very Good"
```

Part B. Constructing Dummy Variables

1. Construct your dummy-variable coding table.

```r
knitr::kable(
  data.frame(
    Cut = c("Fair", "Good", "Very Good", "Premium", "Ideal"),
    Fair = c(1, 0, 0, 0, 0),
    Good = c(0, 1, 0, 0, 0),
    VeryGood = c(0, 0, 1, 0, 0),
    Premium = c(0, 0, 0, 1, 0)
  ),
  caption = "Dummy-variable coding for cut"
)
```

Table 1: Dummy-variable coding for cut

Cut	Fair	Good	VeryGood	Premium
Fair	1	0	0	0
Good	0	1	0	0
Very Good	0	0	1	0
Premium	0	0	0	1
Ideal	0	0	0	0

2. Explain what the all-zero category represents.

The all-zero category represents the reference category, which is Ideal. For an Ideal-cut diamond, all four dummy variables are 0.

3. Explain why five dummy variables should not be included together with an intercept.

Five dummy variables should not be included with an intercept because it would cause perfect multicollinearity. One category must be left out as the reference category, which in this case is Ideal.

Part C. Additive Dummy-Variable Regression

```r
model1 <- lm(price ~ carat + cut, data = diamonds)
summary(model1)
```

```
##
## Call:
## lm(formula = price ~ carat + cut, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17540.7   -791.6    -37.6    522.1  12721.4
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  -2074.55      14.23 -145.77   <2e-16 ***
## carat         7871.08      13.98  563.04   <2e-16 ***
## cutFair      -1800.92      39.34  -45.77   <2e-16 ***
## cutGood       -680.59      24.00  -28.36   <2e-16 ***
## cutPremium    -361.85      16.69  -21.68   <2e-16 ***
## cutVery Good  -290.79      17.24  -16.87   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1511 on 53934 degrees of freedom
## Multiple R-squared:  0.8565, Adjusted R-squared:  0.8565
## F-statistic: 6.437e+04 on 5 and 53934 DF,  p-value: < 2.2e-16
```

1. Write the estimated regression equation from the R output.

```
coef(model1)
```

```
##  (Intercept)        carat      cutFair      cutGood   cutPremium cutVery Good
##   -2074.5457    7871.0821   -1800.9240    -680.5921    -361.8468    -290.7886
```

2. Interpret the coefficient of carat in the context of diamond price.

```
coef(model1)["carat"]
```

```
##    carat
## 7871.082
```

Holding cut constant, a one-carat increase in diamond weight is associated with an increase in the predicted price equal to the estimated carat coefficient.

3. Interpret one dummy-variable coefficient relative to the reference category.

```
coef(model1)["cutFair"]
```

```
##   cutFair
## -1800.924
```

Holding carat constant, the coefficient for Fair represents the difference in predicted price between a Fair-cut diamond and an Ideal-cut diamond.

4. Interpret the R2 of the model.

```
summary(model1)$r.squared
```

```
## [1] 0.8564748
```

The R-squared value represents the proportion of variation in diamond price explained by carat and cut in the additive regression model.

5. Based on the additive model, do the regression lines for the different cut categories have the same slope or different slopes? Explain.

They have the same slope. The additive model assumes that the effect of carat on price is the same for all cut categories; only the intercepts differ.

Part D. Interaction Between Carat and Cut

```r
model2 <- lm(price ~ carat * cut, data = diamonds)

summary(model2)
```

```
##
## Call:
## lm(formula = price ~ carat * cut, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14878.3   -793.0    -23.0    546.3  12706.2
##
## Coefficients:
##                     Estimate Std. Error  t value Pr(>|t|)
## (Intercept)        -2300.37      19.46 -118.234  < 2e-16 ***
## carat               8192.39      23.57  347.568  < 2e-16 ***
## cutFair              461.30      86.57    5.329 9.93e-08 ***
## cutGood             -122.35      49.35   -2.479 0.013170 *
## cutPremium           -79.53      32.07   -2.480 0.013155 *
## cutVery Good        -117.29      33.71   -3.479 0.000503 ***
## carat:cutFair      -2267.90      76.05  -29.820  < 2e-16 ***
## carat:cutGood       -712.76      52.67  -13.533  < 2e-16 ***
## carat:cutPremium    -384.64      34.18  -11.253  < 2e-16 ***
## carat:cutVery Good  -256.42      37.89   -6.768 1.32e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1498 on 53930 degrees of freedom
## Multiple R-squared:  0.8591, Adjusted R-squared:  0.859
## F-statistic: 3.653e+04 on 9 and 53930 DF,  p-value: < 2.2e-16
```

1. State the null and alternative hypotheses for the overall carat x cut interaction.

Null hypothesis: There is no interaction between carat and cut. The effect of carat on price is the same across cut categories.

Alternative hypothesis: There is an interaction between carat and cut. The effect of carat on price differs across cut categories.

2. Using the regression output, identify the interaction coefficients and their corresponding p-values.

```r
summary(model2)$coefficients
```

```
##                       Estimate Std. Error     t value      Pr(>|t|)
## (Intercept)        -2300.37369   19.45616 -118.233709  0.000000e+00
## carat               8192.39130   23.57062  347.567927  0.000000e+00
## cutFair              461.30001   86.56860    5.328722  9.930208e-08
## cutGood             -122.35436   49.35212   -2.479212  1.317034e-02
## cutPremium           -79.53128   32.07376   -2.479637  1.315466e-02
## cutVery Good        -117.28656   33.70895   -3.479389  5.029550e-04
## carat:cutFair      -2267.89620   76.05341  -29.819785 8.194591e-194
## carat:cutGood       -712.75500   52.66960  -13.532568  1.175051e-41
## carat:cutPremium    -384.63927   34.18104  -11.253002  2.413692e-29
## carat:cutVery Good  -256.41934   37.88583   -6.768213  1.317118e-11
```

The interaction terms are examined to determine whether the slope of carat differs from the reference category, Ideal.

3. Use an appropriate ANOVA/incremental F -test to determine whether the interaction contributes significantly to the model.

```
anova(model1, model2)
```

```
## Analysis of Variance Table
##
## Model 1: price ~ carat + cut
## Model 2: price ~ carat * cut
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1  53934 1.2321e+11
## 2  53930 1.2099e+11  4 2223124519 247.73 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The interaction contributes significantly to the model, $F(4, 53930) = 247.735,\ p < .001$. Since $p < .05$, we reject H_0. Therefore, the effect of carat on price differs across the different cut categories.

4. At alpha = 0.05, state your statistical decision and explain what it means in the context of diamond prices.

```
anova(model1, model2)
```

```
## Analysis of Variance Table
##
## Model 1: price ~ carat + cut
## Model 2: price ~ carat * cut
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1  53934 1.2321e+11
## 2  53930 1.2099e+11  4 2223124519 247.73 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Using a significance level of 0.05, reject the null hypothesis if the p-value is less than 0.05.

5. Explain the practical difference between the additive model and the interaction model.

The additive model assumes that all cut categories have the same slope for carat but different intercepts.

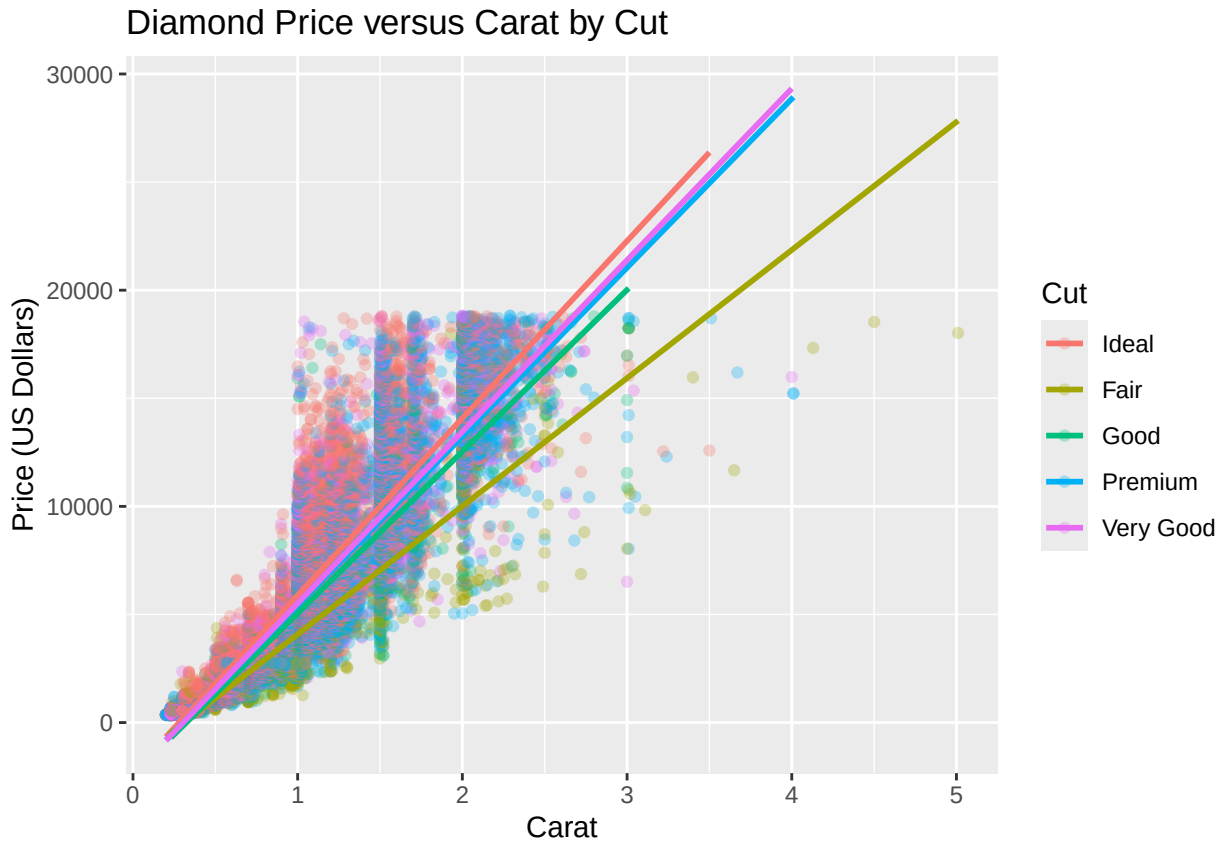
The interaction model allows both the intercepts and the slopes to differ across cut categories.

Part E. Visualization and Statistical Conclusion

```
ggplot(
  diamonds,
  aes(x = carat, y = price, color = cut)
) +
  geom_point(alpha = 0.3) +
  geom_smooth(
    method = "lm",
    se = FALSE
  ) +
```

```
  labs(
    title = "Diamond Price versus Carat by Cut",
    x = "Carat",
    y = "Price (US Dollars)",
    color = "Cut"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The graph shows a positive relationship between carat and diamond price, indicating that diamonds with greater carat generally have higher prices. The cut categories also show differences in their relationship with price. The additive model assumes that the effect of carat is the same across all cut categories, while the interaction model allows the effect of carat to differ by cut. The incremental F-test is used to determine whether adding the interaction terms significantly improves the model. Based on the resulting p-value, the appropriate statistical conclusion is reported above.